

- Q.22 Explain one type of earth return system in automobile with diagram.
- Q.23 Explain the construction of Lithium ion battery.
- Q.24 Describe trickle charging method.
- Q.25 Describe Cut-Out Relay used in Automobiles.
- Q.26 Discuss five advantages of dynamo and alternators.
- Q.27 Discuss the head lamp and vertical and side control of it.
- Q.28 Differentiate between HT and LT cables.
- Q.29 Write any five cable colour codes.
- Q.30 Describe the significance of wiring harness.
- Q.31 Enlist various sensors used in automobiles? Explain any one in detail.
- Q.32 Explain the hybrid vehicles with diagram.
- Q.33 Write the construction of fuel cell with diagram.
- Q.34 Describe the working of Defogger used in Automobiles.
- Q.35 Discuss the foldable ORVM.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Describe various methods of battery charging.
- Q.37 Explain the principle and construction of Starter motor with neat sketch.
- Q.38 Explain the working principle, construction and working of lead acid battery with diagram.

(80) (4) 180355/170355/
120355/030355

No. of Printed Pages : 4
Roll No.

180355/170355/
120355/030355

5th Sem / Auto Sub : Auto Electrical & Electronics Systems

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 No of cells in a 24 volt battery is
a) 2 b) 4
c) 6 d) 12
- Q.2 Cells are connected in series because
a) Increase in voltage rating
b) Increase in current rating
c) Increase the life of the cell
d) None of the above
- Q.3 The automotive battery is also known as ____
a) Lithium ion battery
b) Lead-acid storage battery
c) Zinc carbon battery
d) Weston cell battery
- Q.4 The instrument used to check specific gravity of acid in a battery is _____.
a) Hydrometer b) Hygrometer
c) Anemometer d) Multi-meter

(1) 180355/170355/
120355/030355

- Q.5 What is the colour of a positive plate of a lead-acid battery?
- a) White b) Brown
c) Gray d) Black
- Q.6 The purpose of the dynamo in an Automobile is to
- a) Supply electric power
b) Convert mechanical energy into electrical energy
c) Continually recharge the battery
d) None of the above
- Q.7 The brake warning light warns the driver of _____
- a) Water in the master cylinder
b) Air in the hydraulic system
c) Failure of the primary or secondary circuit of hydraulic system
d) Power brake failure
- Q.8 The standard E.M.F. of the hydrogen-oxygen fuel cell is _____.
- a) 1.23 v b) 2.34 v
c) 3.45 v d) 3.55 v
- Q.9 Which of the following is not an example of a fuel cell?
- a) Hydrogen-oxygen cell
b) Methyl-oxygen-alcohol cell
c) Propane-oxygen cell
d) Hexanone oxygen cell

(2)

180355/170355/
120355/030355

- Q.10 ECU stands for
- a) electronic control unit
b) Electric control unit
c) Both A & B
d) None of the above

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Symbol of Fan used in Automobiles is _____.
- Q.12 Write the function of Horn in Automobiles.
- Q.13 Electrolyte use in alkaline battery is _____.
- Q.14 Battery voltage testing is done by _____.
- Q.15 Red colour wire in Automobiles indicates _____ wire.
- Q.16 Full form of LED _____.
- Q.17 Write the function of Reversing Light in Automobile.
- Q.18 Define fuel cell.
- Q.19 Write the purpose of odometer in an automobile?
- Q.20 Write name of any two types of fuel cell.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Make a list of ten electrical components that are used in automobile.

(3)

180355/170355/
120355/030355